TECHTONIC UI 3.0 · Hackathon Competition
STASH · Smart Technology and Security Hackathon

SentriFlow

A Local Network Privacy Guardian for Real-Time Visibility and Before-Forward Control

Visibility. Understanding. A decision before forwarding.

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Full Proposal · Working Draft

Prepared for review. User research, independent review and official submission requirements remain to be confirmed.

Introduction

Laptop applications send network requests during everyday work, while their owners may struggle to identify which application contacted a destination or whether a transfer includes sensitive content. Encryption protects traffic in transit but also means an authorized local content-inspection tool needs explicit trust configuration and careful handling of plaintext. SentriFlow asks whether a local interface can help a device owner understand and stop a potentially sensitive HTTP request before it is forwarded.

The intended users are students, remote workers, privacy-conscious laptop owners and small teams without a dedicated network analyst. These are target groups, not groups already validated by a survey. Prototype targets are at least 90% named-process attribution during a documented representative capture session, detection of all seeded dummy PII fixtures, and a local decision applied within one second. Targets must be reported separately from actual measurements.

Problem Analysis

The design hypothesis is that connection metadata alone can leave users unsure about the contents and significance of a transfer, while detailed protocol inspection may require specialist knowledge. The product must connect the process, destination, a bounded content signal and an actionable decision in one workflow. Root causes to investigate include background application behavior, encrypted content, unfamiliar terminology, coarse persistent permissions and the effort of using expert tools.

Application firewalls and protocol analyzers already address important parts of this problem. mitmproxy itself supplies programmable HTTP(S) interception [1]. SentriFlow composes existing techniques into an interface around redacted pattern signals and per-request decisions; it does not claim its interception mechanism is novel or that other tools categorically lack similar capabilities. User-validation questions and competitor comparisons need primary evidence and observed tasks before stronger claims are justified.

Proposed Solution

The owner explicitly routes a client to a loopback proxy. SentriFlow identifies the client socket's process where possible, scans supported representations for email, Indonesian-phone heuristics, Luhn-valid card candidates and locally configured sensitive strings, and checks local tracker domains. Risky or incompletely inspected requests enter Halt. The dashboard displays process, host, size and redacted pattern labels. The user can Allow/Block once or remember a process/exact-host decision; silence defaults to Block.

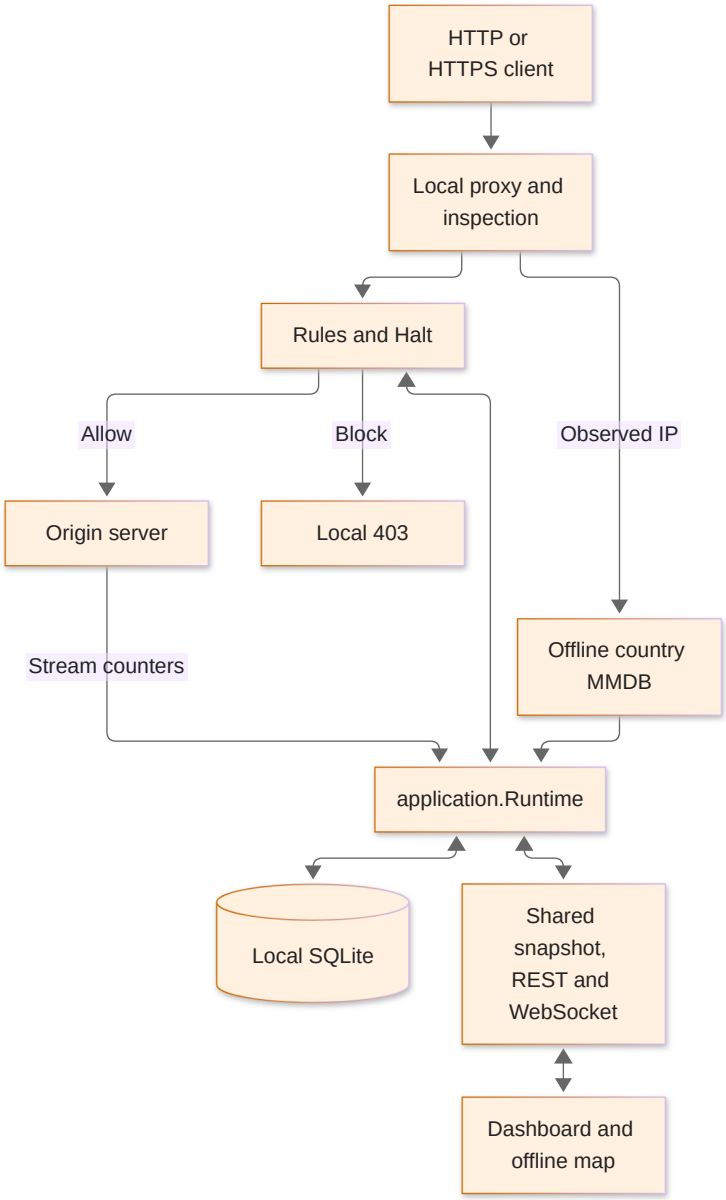
In the local demonstration a client submits a dummy email, phone and test card to a local origin. The real dashboard presents Halt before the payload arrives at the origin. Block produces a local 403; Allow reaches the origin. The history, alerts and disclosed score update. The privacy score starts at 100, subtracts 10 per allowed PII request and 2 per allowed tracker request, adds 5 per blocked PII request and 1 per blocked tracker request, and clamps the result to 0–100. The dashboard discloses the formula; it is not a claim of total device safety.

System Design

The implementation follows the supplied repository build plan: Python package under `backend/src/sentriflow`, React pages/hooks/stores, loopback proxy 8080 and dashboard/API 8787. Domain logic has no dependency on API/proxy/Windows adapters. A single asyncio process shares the decision coordinator between mitmproxy and FastAPI. Each hold has its own Future and authoritative deadline. WebSocket events communicate flows, holds, scores and a one-second speed ticker.

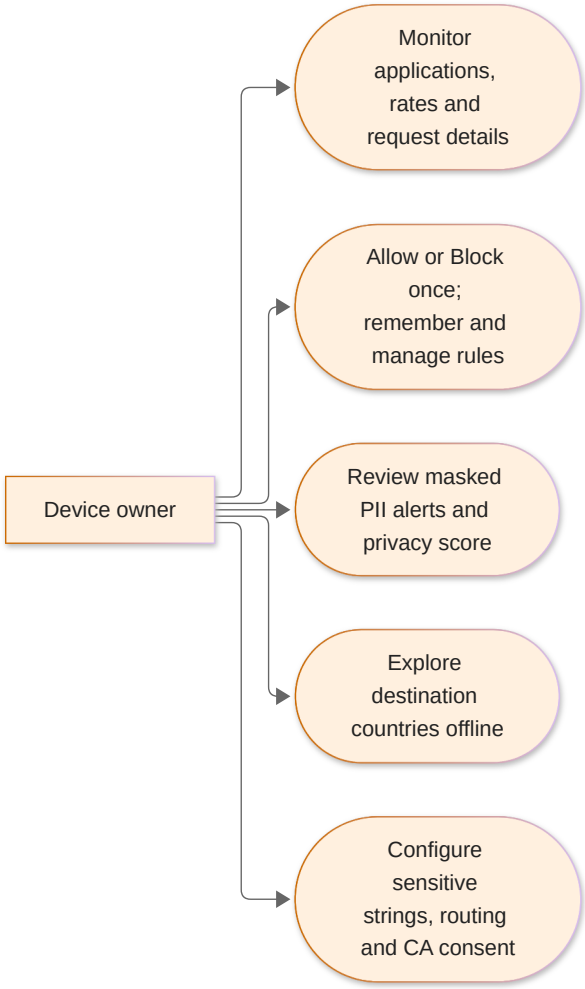
Architecture, use-case, Halt, ERD and API diagrams are maintained in `diagrams/`. The API contract and architecture document actual names and deliberate deviations. SQLite stores redacted records, rules, alerts, source domains and public settings. Raw payloads/headers/URL queries are excluded. Sensitive literal/regex definitions are persisted locally for future matching; public snapshots omit their values. Raw request content remains in memory only. The API, inspection and proxy share application.Runtime; the runner owns signal handling, restoration and shutdown.

Figure 1. System architecture



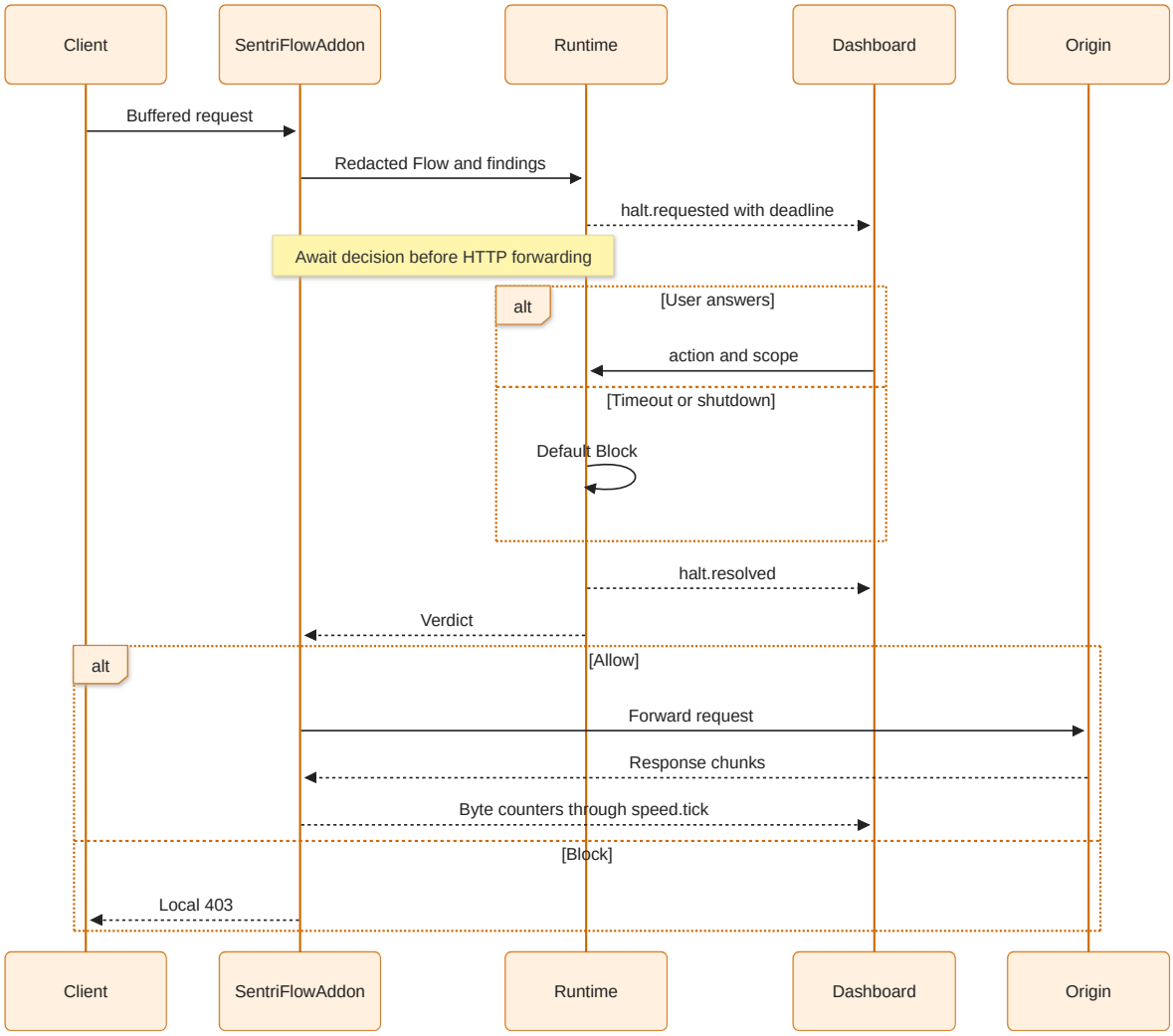
The control plane and proxy share a single Runtime. Country lookup uses a local database and the destination IP observed on the upstream socket; it does not resolve captured hostnames just to draw the map.

Figure 2. Owner use cases



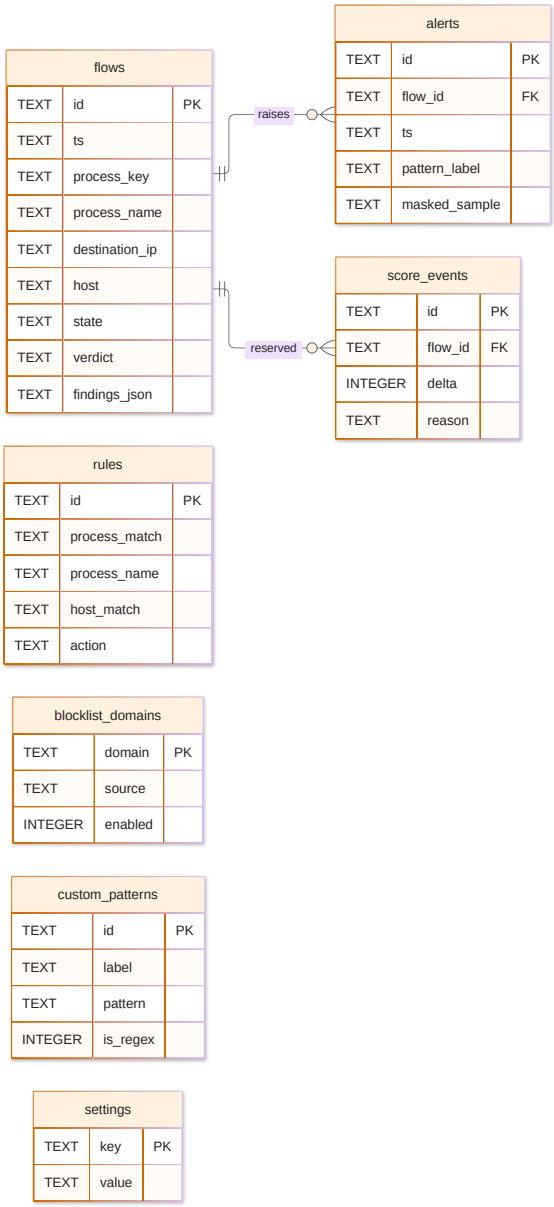
The owner can inspect traffic details, manage exact-scope rules, review masked alerts, explore destination countries and explicitly consent to routing or CA trust changes.

Figure 3. Halt sequence



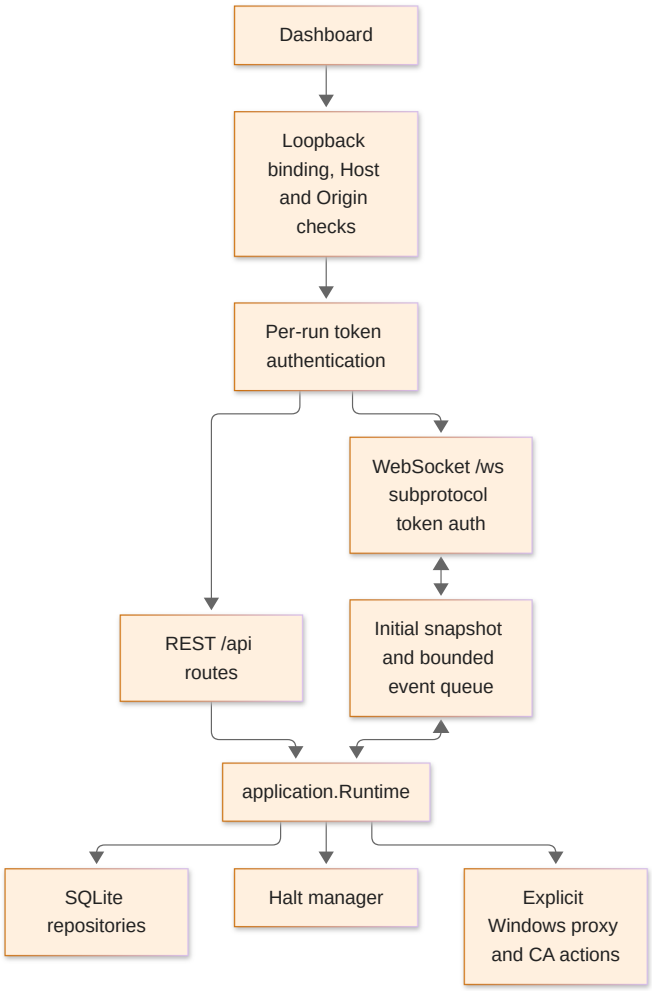
The authoritative deadline belongs to the backend. The interface countdown explains it; a missing or disconnected interface never turns an unanswered hold into Allow.

Figure 4. Database relationships



Relational flow records include observed destination IP, optional country code, lifecycle state and masked findings JSON. Migration 002 preserves schema-v1 history/rules while adding dashboard fields. The custom-pattern table stores user definitions; process-baseline and score-event tables remain reserved, and the anomaly baseline is not implemented.

Figure 5. API and event boundary



Route	Purpose
`GET /api/snapshot`	Redacted dashboard state and pending holds
`GET /api/flows`, `/processes`, `/alerts`, `/score`	Retained request details, session rates and activity indicators
`POST /api/halts/{id}/decision`	Allow/Block once or always
`GET/POST/PUT/DELETE /api/rules`	Manage persistent process/exact-host rules
`GET /api/geo/destinations`	Country aggregation from a local MMDB, with unplotted counts
`GET/PUT /api/settings`, `GET/POST/DELETE /api/patterns`	Inspection configuration and locally stored patterns
`/ws`	Subprotocol token authentication, typed events and an initial snapshot

DELETE routes use resource IDs. Full request/response shapes and the explicit Windows consent routes are defined in the API document.

Technical Implementation

mitmproxy provides authorized HTTP/HTTPS interception [1]. The asynchronous request hook waits for the verdict before forwarding the request. Response chunks are counted as they arrive, so download rates can update before response completion [2]; outbound request streaming remains disabled to preserve inspection-before-forward. psutil maps the loopback client endpoint to PID/executable identity, returning Unknown if permissions or process lifetime prevent reliable attribution.

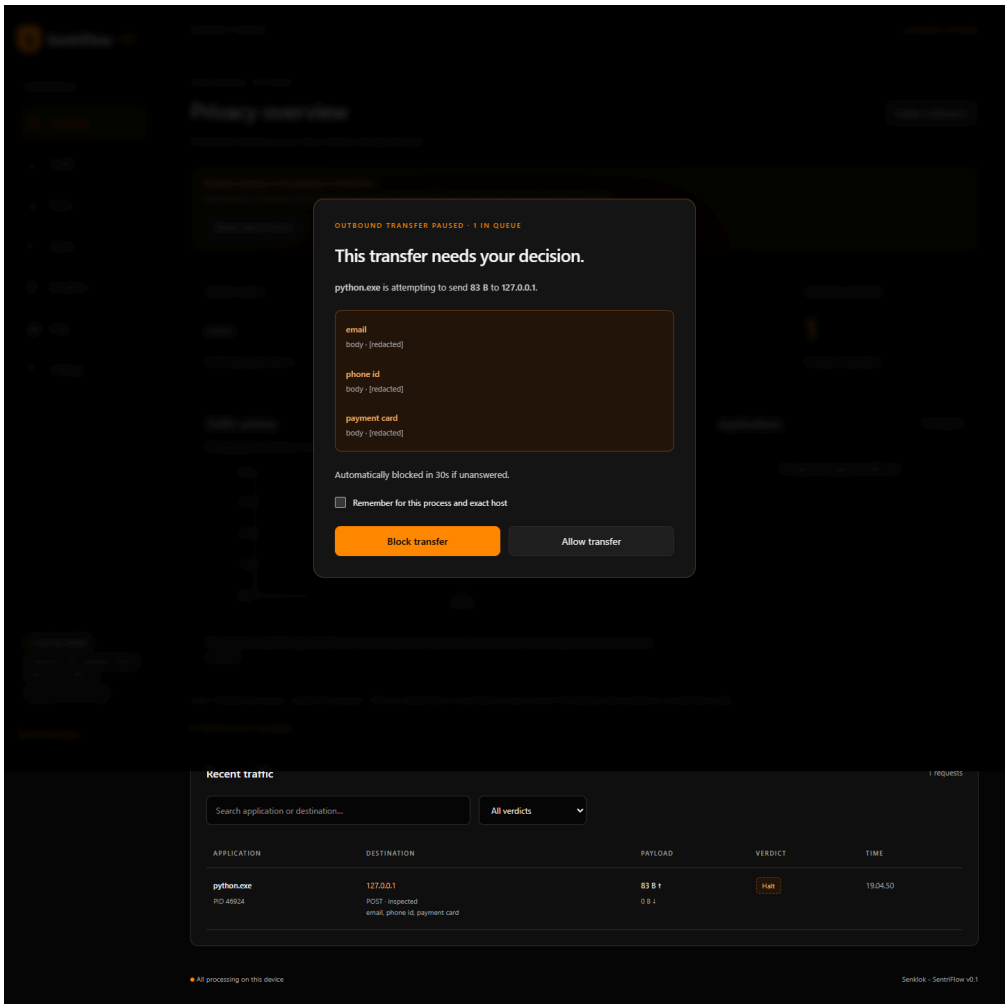
Inspection is bounded to 1 MiB of decoded content; the proxy rejects request bodies above 2 MiB. Supported UTF-8 text/JSON/form/XML, gzip/deflate and multipart text fields are inspected. Unsupported encodings, binary/invalid text and decoding limits prompt as incomplete. Custom regex execution has a timeout. R3's hosts/EasyPrivacy domain subset loads local reviewed files and caches the matcher between updates; no threat feed is silently downloaded. The behavioral baseline remains future work.

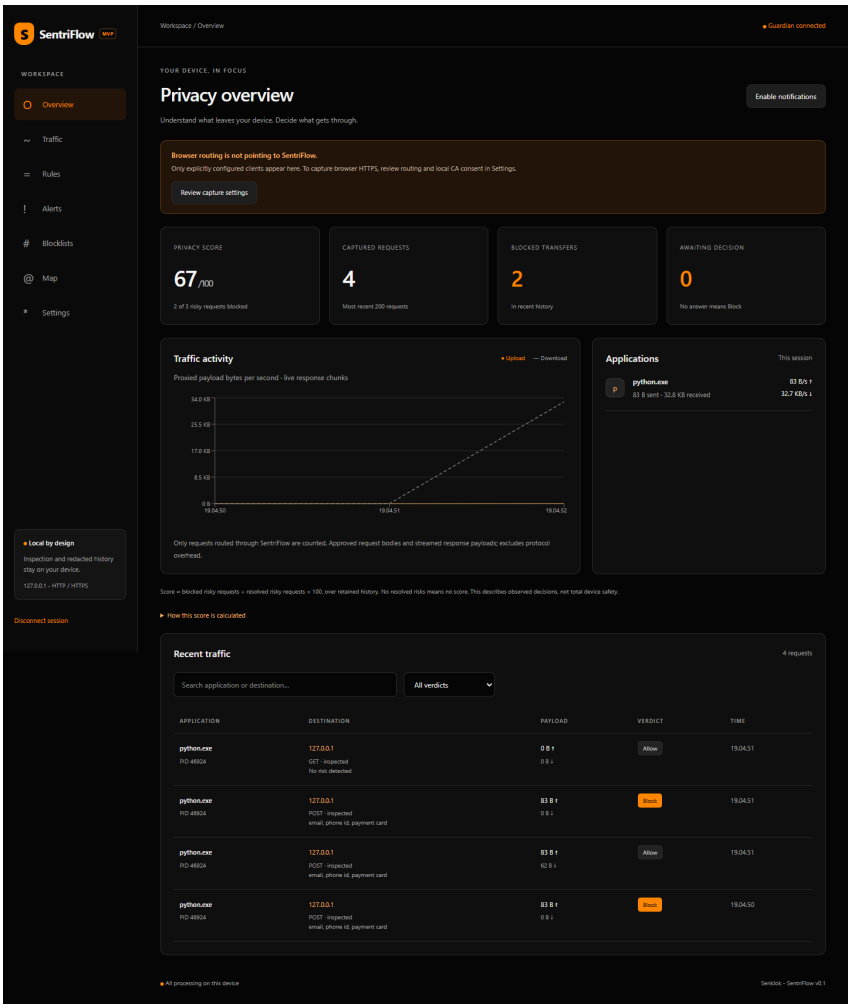
The R4 interface provides upload/download rates per application, searchable traffic and redacted request details, the queued Halt prompt, rule editing/deletion, severity-filtered alerts and a disclosed privacy-score breakdown. The responsive black-and-papaya-orange layout uses local assets only. An unsuccessful decision remains visible inside the Halt dialog, and uninspected requests cannot be remembered as Allow.

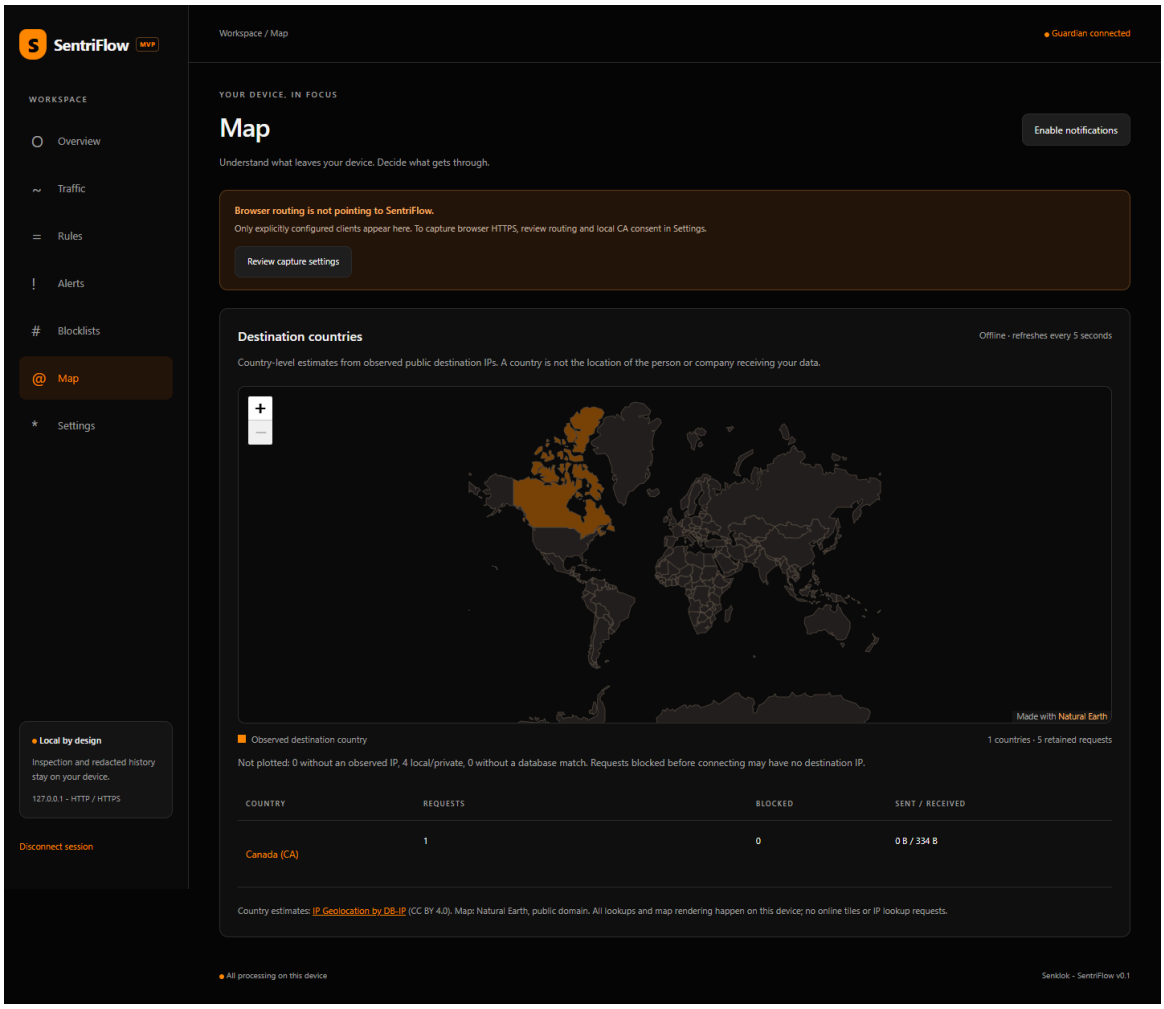
Destination countries are rendered with Leaflet [4] and bundled Natural Earth shapes [5]. Country estimates are read from a local DB-IP Lite country MMDB [6]. The map groups retained requests, distinguishes blocked transfers and payload sizes, and accounts for private, unobserved or unmatched IPs. It never contacts an online tile server or an IP lookup service. These approximate IP-country assignments are not evidence of a recipient's nationality, identity or physical location. DB-IP attribution and data provenance are included with the map.

The service binds to loopback and requires a per-run session token with accepted Host/Origin. WebSocket authentication uses the sentriflow subprotocol handshake rather than a query parameter [3]. Its first status event contains the dashboard snapshot, followed by queued changes to avoid reconnect races. CA keys are generated per private data directory; trust install is explicit. Upstream certificate verification remains enabled. Windows routing is opt-in with a durable original-settings journal, normal-exit restore and a hidden watchdog for dead-session recovery. Fake-registry tests validate the logic; real Windows recovery acceptance remains necessary.

Local tests cover PII/redaction, Luhn, domain boundaries, rule scope, timeout/parallel holds, persistence/recovery, API/WS access and real HTTP/HTTPS non-delivery/release. A dedicated test confirms positive download counter/rate while the origin deliberately holds the remaining response. Historical Edge UI checks and screenshots are linked below. The integrated four-role app also has a reproducible local browser check covering reconnects, remembered rules, settings, streaming and offline rendering; see the current integration status document.


OUTBOUND TRANSFER PAUSED · 1 IN QUEUE
This transfer needs your decision.
python.exe is attempting to send 83 B to 127.0.0.1.
email
body · [redacted]
phone id
body · [redacted]
payment card
body · [redacted]
Automatically blocked in 30s if unanswered.
Remember for this process and exact host
Block transfer
Allow transfer
Recent traffic
Search application or destination...
All verdicts
APPLICATION
DESTINATION
PAYLOAD
VERDICT
TIME
python.exe
PID 46924
127.0.0.1
POST · inspected
email, phone id, payment card
83 B ↑
0 B ↓
Halt
19.04.50
All processing on this device
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SentriFlow
MVP
WORKSPACE
Overview
Traffic
Rules
Alerts
Blocklists
Map
Settings
Local by design
Inspection and redacted history stay on your device.
127.0.0.1 - HTTP / HTTPS
Disconnect session
Workspace / Overview
Guardian connected
YOUR DEVICE, IN FOCUS
Privacy overview
Understand what leaves your device. Decide what gets through.
Enable notifications
Browser routing is not pointing to SentriFlow.
Only explicitly configured clients appear here. To capture browser HTTPS, review routing and local CA consent in Settings.
Review capture settings
PRIVACY SCORE
67 /100
2 of 3 risky requests blocked
CAPTURED REQUESTS
4
Most recent 200 requests
BLOCKED TRANSFERS
2
In recent history
AWAITING DECISION
0
No answer means Block
Traffic activity
Upload
Download
Proxied payload bytes per second - live response chunks
34.0 KB
25.5 KB
17.0 KB
8.5 KB
0 B
19:04:50
19:04:51
19:04:52
Only requests routed through SentriFlow are counted. Approved request bodies and streamed response payloads; excludes protocol overhead.
Applications
This session
python.exe
83 B sent - 32.8 KB received
83 B/s ↑
32.7 KB/s ↓
Score = blocked risky requests ÷ resolved risky requests × 100, over retained history. No resolved risks means no score. This describes observed decisions, not total device safety.
How this score is calculated
Recent traffic
4 requests
Search application or destination...
All verdicts
APPLICATION
DESTINATION
PAYLOAD
VERDICT
TIME
python.exe
PID 46924
127.0.0.1
GET · inspected
No risk detected
0 B ↑
0 B ↓
Allow
19:04:51
python.exe
PID 46924
127.0.0.1
POST · inspected
email, phone id, payment card
83 B ↑
0 B ↓
Block
19:04:51
python.exe
PID 46924
127.0.0.1
POST · inspected
email, phone id, payment card
83 B ↑
62 B ↓
Allow
19:04:31
python.exe
PID 46924
127.0.0.1
POST · inspected
email, phone id, payment card
83 B ↑
0 B ↓
Block
19:04:30
All processing on this device
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The map screenshot comes from an explicit non-personal test request to example.com through the proxy. Geography uses the observed server IP and a local database; the browser itself made no external tile or lookup requests. IP routing and dataset changes can alter the country shown in a future run.

Measured evidence is in local-demo-report.json: three local dummy requests, detected pattern labels, named attribution and click-to-response timing. This small controlled sample establishes feasibility; it does not demonstrate representative user behavior, universal detection or a production safety guarantee.

Expected MVP, Impact and Plan

Core capture/inspection/Halt/rules/history/dashboard and the offline destination map are implemented. Remaining acceptance work is a clean Windows 11 installation, representative multi-app session, real normal/forced-crash routing restoration, detector false-positive review, compatibility/pinning/QUIC tests, long-run stability and notification delivery. Anomaly work follows core reliability. Kernel filtering, mobile support, installer and a trained PII model remain out of scope.

R4 deliverable	Implementation and review evidence
Dashboard and traffic	React/Zustand, real WebSocket events, per-application rates, filters and request detail

R4 deliverable	Implementation and review evidence
Halt and rules	Countdown, queued holds, Allow/Block once/always, rule editing/deletion, visible decision errors
Alerts and score	High-priority PII labels, redaction, severity filter and disclosed weighted score calculation
Offline country map	Local MMDB plus bundled vector countries, explicit unknown-IP accounting and accessible country table
Proposal package	Consistent cover, five exported diagram sets, local HTML and A4 PDF working draft

Impact hypotheses are improved user comprehension of application/destination/pattern signals and the ability to prevent a demonstrated dummy transfer. Measure task success, explanation accuracy, decision time, unanswered prompts and disruption with consented participants. Do not claim breach reductions or legal compliance from this prototype. Research data and quotes have not yet been collected.

The remaining plan is acceptance/stabilization first, consented validation and source review second, then independent proposal review, rehearsal and backup video. A reviewable draft PDF and diagram exports are generated from local sources. Dates, rubric weights and the suggested 15-page cap from the tracker are planning inputs until confirmed by official organizer material. The draft has not been submitted; user-validation findings and official-rule review must precede a submission-ready version.

Risk	Mitigation / boundary
Proxy bypass / QUIC	Document capture scope; test selected browser routing and fallback before demo
Trust/pinning failure	Mark uninspected; no pinning bypass or blanket TLS verification disable
False positives / prompt fatigue	Bounded inspection, transparent labels, exact-scope rules, participant testing
Unanswered prompt	Short authoritative timeout and default Block
CA or activity metadata exposure	Restricted local directory, redaction, explicit trust removal
Broken system proxy	Durable snapshot, ordered shutdown/restore, watchdog and emergency command
Demo/network failure	Offline origin, automated local tests, genuine screenshots; backup recording pending

References

[1] mitmproxy contributors, “Event Hooks,” mitmproxy Documentation. [Online]. Available: https://docs.mitmproxy.org/stable/api/events.html. [Accessed: Oct. 27, 2026].

[2] mitmproxy contributors, “HTTP API,” mitmproxy Documentation. [Online]. Available: https://docs.mitmproxy.org/stable/api/mitmproxy/http.html. [Accessed: Oct. 27, 2026].

[3] FastAPI, “WebSockets,” FastAPI Documentation. [Online]. Available: https://fastapi.tiangolo.com/advanced/websockets/. [Accessed: Oct. 27, 2026].

[4] Leaflet contributors, “API Reference,” Leaflet 1.9.4. [Online]. Available: https://leafletjs.com/reference.html. [Accessed: Oct. 27, 2026].

[5] Natural Earth, “Terms of Use.” [Online]. Available: https://www.naturalearthdata.com/about/terms-of-use/. [Accessed: Oct. 27, 2026].

[6] DB-IP, “IP to Country Lite,” October 2026 edition, CC BY 4.0. [Online]. Available: https://db-ip.com/db/download/ip-to-country-lite. [Accessed: Oct. 27, 2026].